

Database Management System

Subject Code : 2114112

Module1

--Shilpa Ingoley

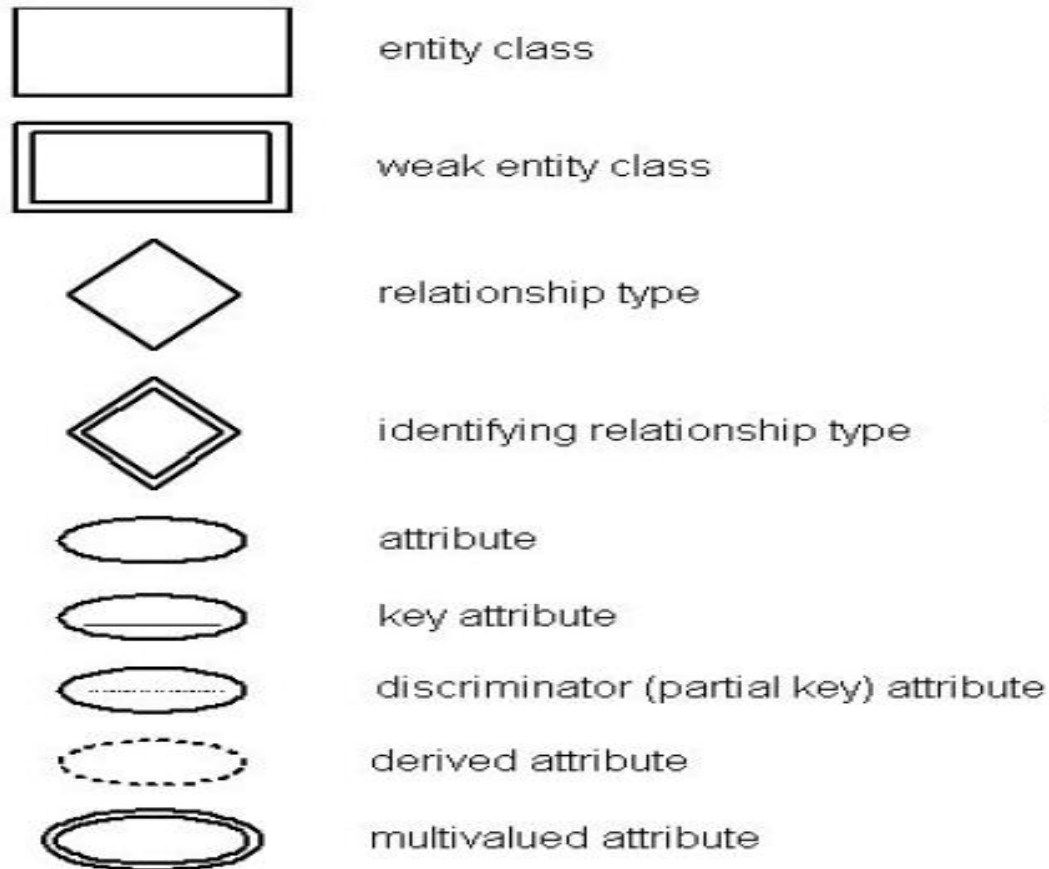
Entity-Relationship (ER) Model

- It is a high-level data model. This model is used to define the data elements and relationship for a specified system.
- It develops a conceptual design for the database. It also develops a very simple and easy to design view of data.
- In ER modeling, the database structure is portrayed as a diagram called an entity-relationship diagram.

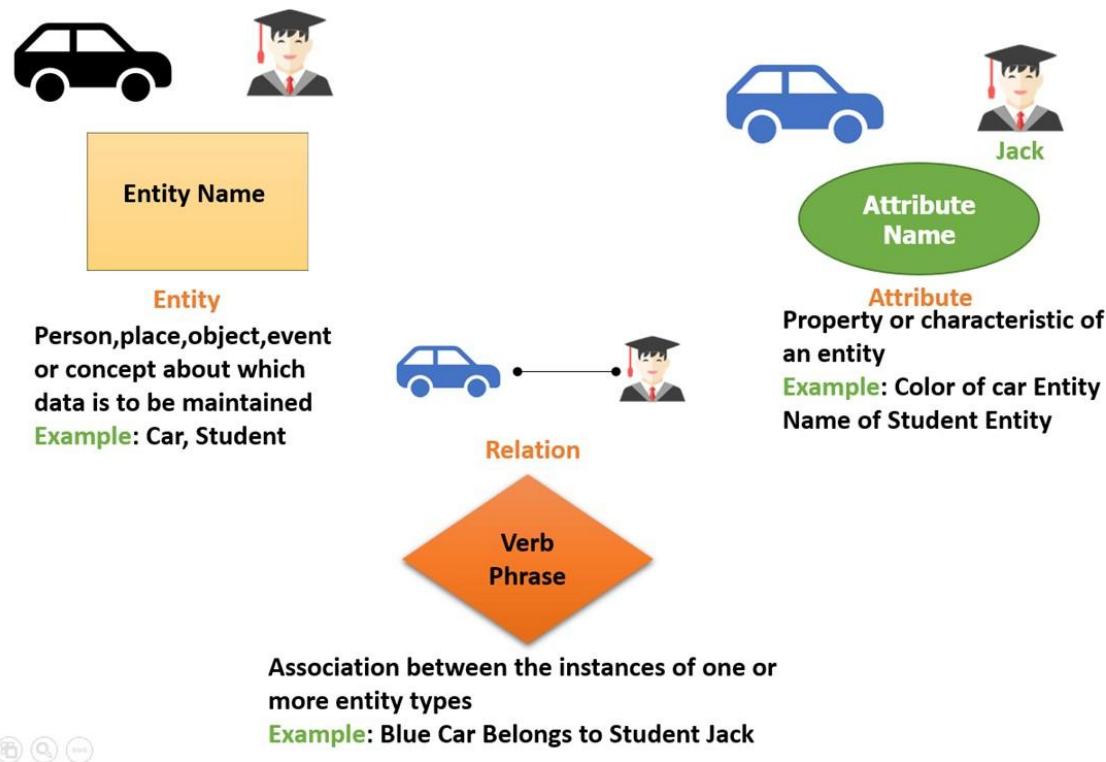
This model is based on three basic concepts:

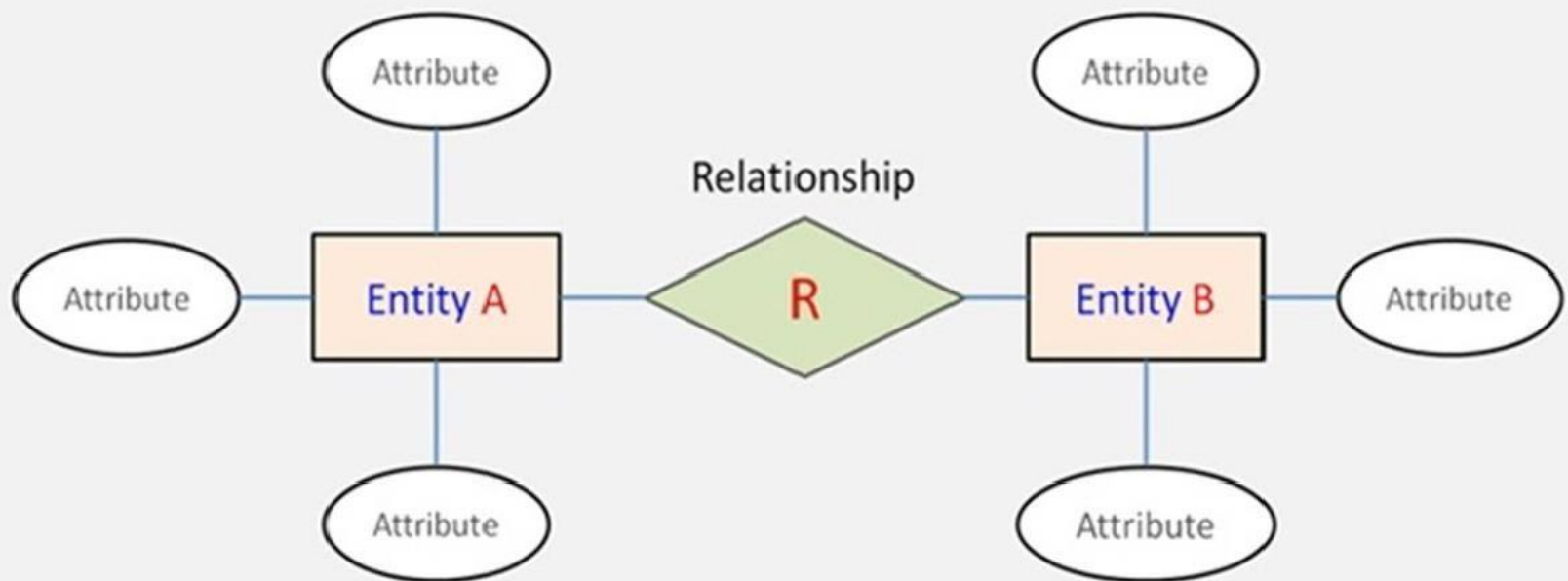
1. Entities
2. Attributes
3. Relationships

Components Of E-R model



For example: in a University database, we might have entities for Students, Courses, and Lecturers. Student entity can have attributes like Rollno, Name, and DeptID. They might have relationships with Courses and Lecturers.





Entity Relationship Diagram (ERD) In DBMS

Entity Relationship Diagram

- It is very much essential to understand the requirement properly and design them efficiently.
- It's a high level Conceptual Model developed by Chen in 1976 to facilitate database design
- Independent of hardware and software constraints
- It is like a foundation of the building(Architectural building plan)
- For this purpose, we use ER diagrams where we plan the database pictorially.
- It represent graphical representation of logical relationship of entities
- ER Data Model is based on the real world objects and their relationship
- *A database* can be modeled as:
 - a collection of entities,
 - relationship among entities.

Entities

- It may be an object with a physical existence – a particular person, car, house, or employee – or it may be an object with a conceptual existence – a company, a job, or a university course.
- Example:

Consider an organization - manager, product, employee, department etc. can be taken as an entity.



- An **entity set** is a set of entities of the same type (e.g., all persons having an account at a bank).

Basic Building Blocks

- ERD comprises of three main components
- 1) Entities/Entity is any thing about which data can be collected and stored
- 2) Attribute is a characteristics of an entity
- 3) Relationship describe the associations among(two or more entities)
 - One-to-one(1:1)
 - One –to-many (1:M)
 - Many-to-many(M:N)

Entity

- An *entity* is an object that exists and is distinguishable from other objects.

Example: an individual student, customer, account

- An *entity set* is a set of entities of the same type that share the same properties.

Example: set of all students, customers, accounts

- Entities are represented by means of rectangles. Rectangles are named with the entity set they represent.



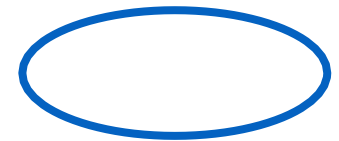
Student

Teacher

Projects

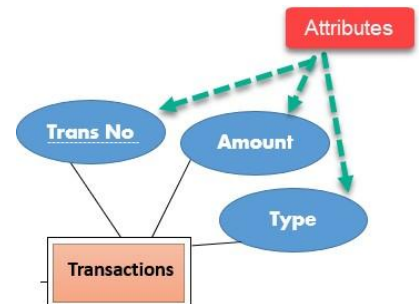
Attributes

- Each entity has a set of associated properties that describes the entity. These properties are known as **attributes**.
- They are represented by an oval.



- **Attributes can be:**

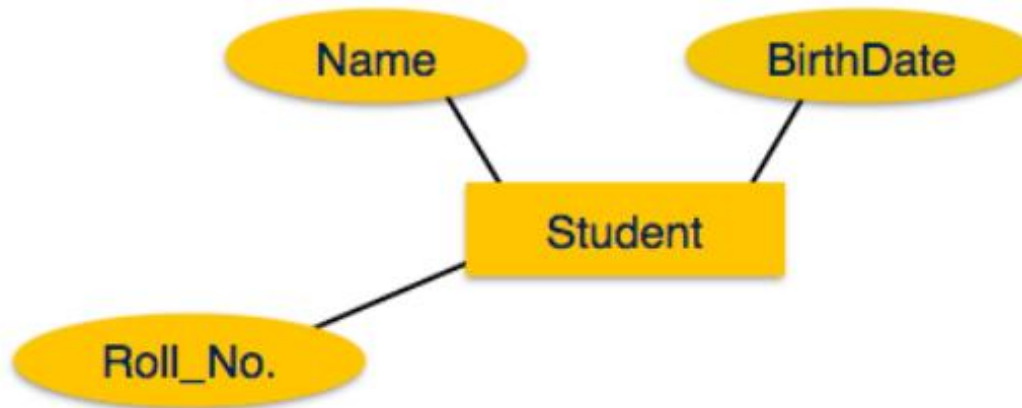
1. Key attribute
2. Simple attribute
3. Composite attribute
4. Single valued attribute
5. Multivalued attribute
6. Stored attribute
7. Derived attribute
8. Complex attribute
9. Null attribute
10. Descriptive attribute



Attributes

■ Entities have *attributes*

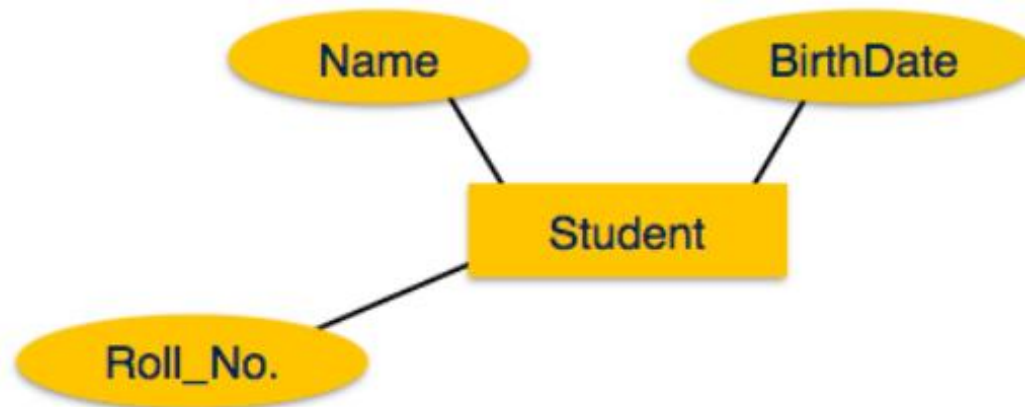
- **Attributes** are the properties of entities. Attributes are represented by means of ellipses. Every ellipse represents one attribute and is directly connected to its entity set (rectangle).
- E.g. Students have *names, Roll number, DOB etc.*
- It's also known as **columns** of the table.



Types of attributes

Depending on the values that an attribute can take, it is divided into different types.

- **Simple Attribute** : These kinds of attributes have values which cannot be divided further.
- Example: STUDENT_ID attribute which cannot be divided into smaller parts.

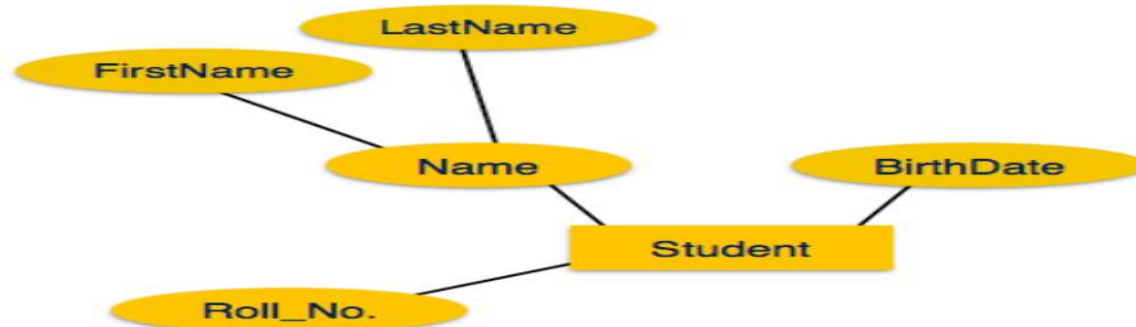


Types of attributes

- **Composite Attribute** : This kind of attribute can be divided further to more than one simple attribute.

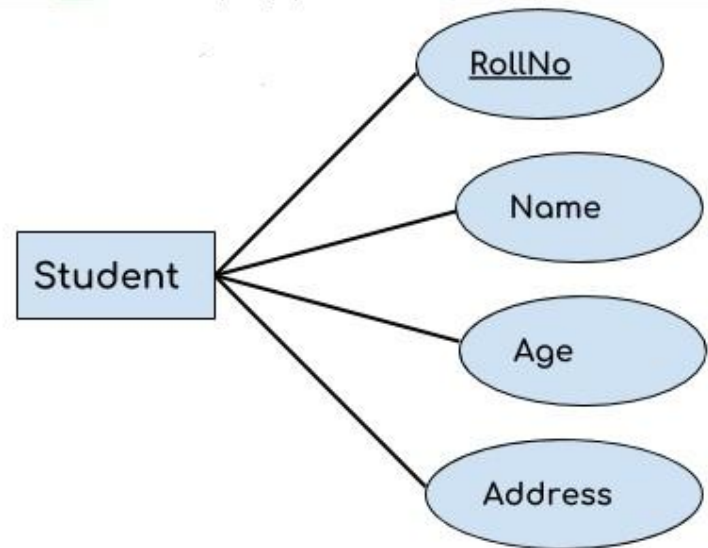
Eg 1: address of a person. Here address can be further divided as Door#, street, city, state and pin which are simple attributes.

Eg 2 : Name



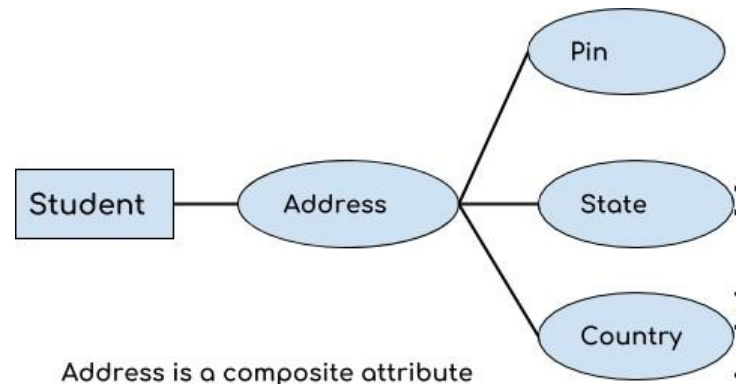
Key attribute

- It can uniquely identify an entity from an entity set.
- **Text of key attribute is underlined.**
- For example, student roll number can uniquely identify a student from a set of students



Composite attribute

- It is a combination of other attributes
- For example, In student entity, the student address is a composite attribute as an address is composed of other attributes such as pin code, state, country.



Composite Attributes

Composite
Attributes

name

first-name middle-initial last-name

Component
Attributes

address

street city state postal-code

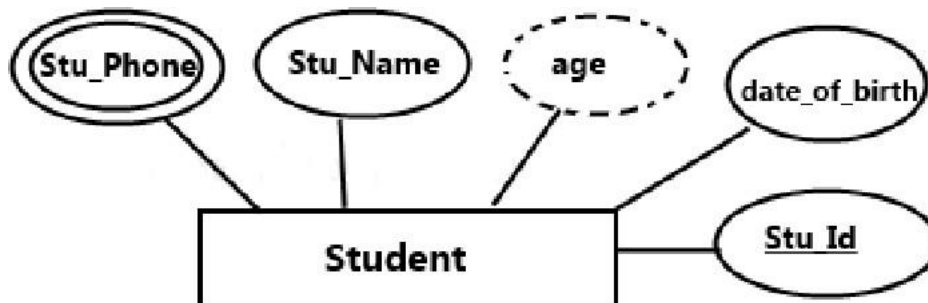
street-number street-name apartment-number

Multivalued attribute

- It can hold multiple values is known as multivalued attribute.
- It is represented with **double ovals** in an ER Diagram.
- For example – A person can have more than one phone numbers so the phone number attribute is multivalued.

Derived attribute:

- It is one whose value is dynamic and derived from another attribute.
- It is represented by **dashed oval** in an ER Diagram.
- For example – Person age is a derived attribute as it changes over time and can be derived from another attribute (Date of birth).



Types of attributes

- **Derived Attribute** : Derived attributes are the one whose value can be obtained from other attributes of entities in the database.

Eg1: Age of a person can be obtained from date of birth and current date.

Eg2: Average salary, annual salary,

Eg3: Total marks of a student etc are few examples of derived attribute.

Derived attributes are depicted by **dashed ellipse**.

- **Stored Attribute**

The attribute which gives the value to get the derived attribute are called Stored Attribute.

Eg: age is derived using Date of Birth. Hence Date of Birth is a stored attribute.

- **Key attribute:** An attribute whose values are distinct for each individual entity in the entity set. Such an attribute is called Key attribute.

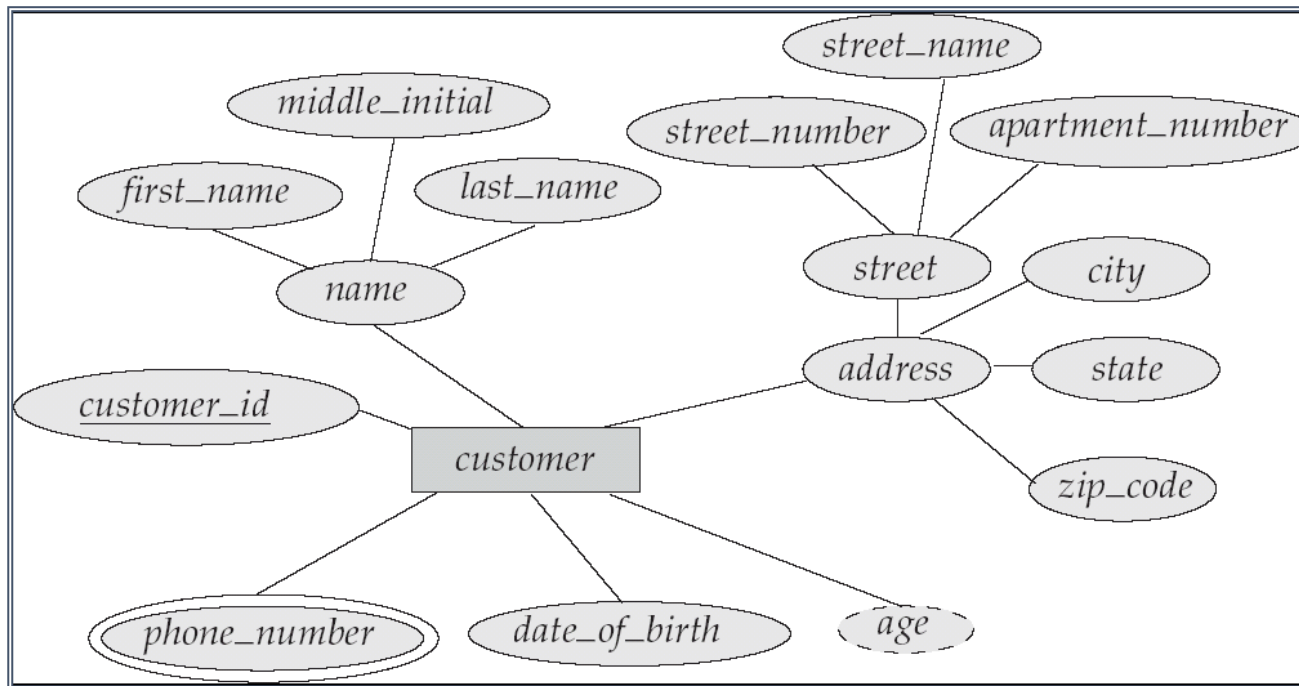
Eg: Employee ID, Student Id, Adhar card

- **Complex attribute:** composite and multivalued attributes can be nested arbitrarily

Eg: Person having multiple addresses

- **Descriptive Attribute:** attributes of relationship set is called descriptive attributes

Customer entity with Composite, Multivalued, and Derived Attributes

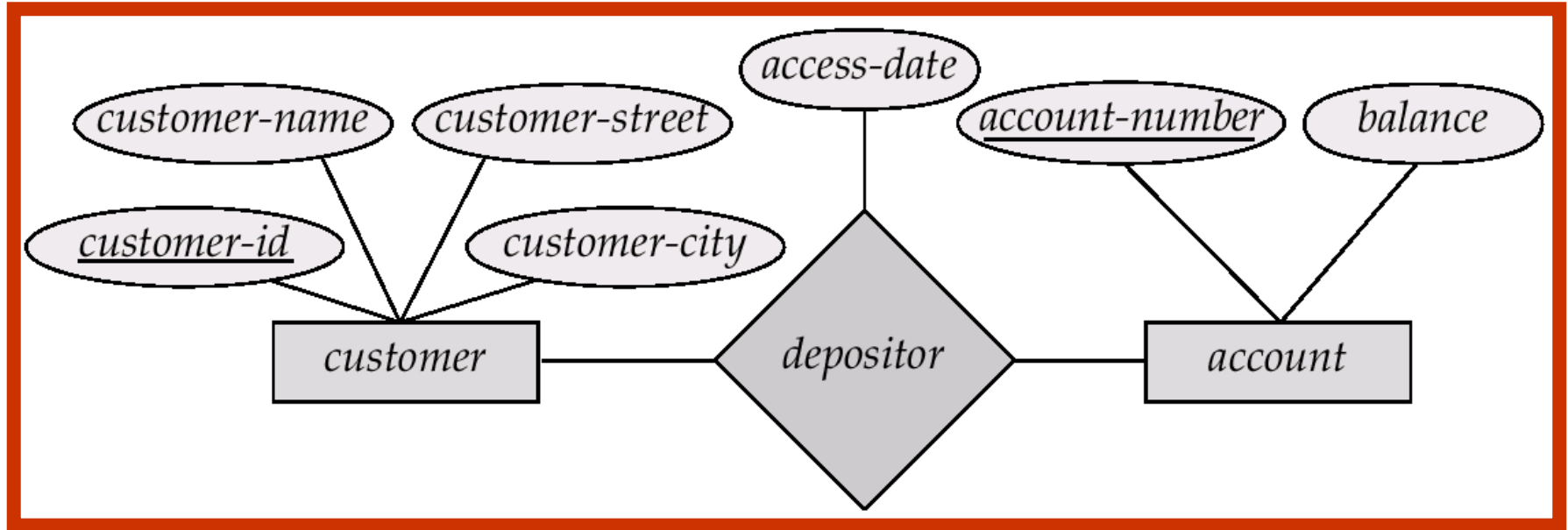


Q) Identify entities and their respective attributes for Library

Relationship

- Relationships are represented by **diamond-shaped** box. Name of the relationship is written inside the diamond-box. All the entities (rectangles) participating in a relationship, are connected to it by a line.

Relationship Sets with Attributes



Questions on E-R Diagrams

- 1) Rectangles represent _____.
- 2) Diamonds represent _____.
- 3) _____ attributes to entity sets and entity sets to relationship sets.
- 4) Ellipses represent _____
- 5) Double ellipses represent _____
- 6) Dashed ellipses denote _____.
- 7) _____ indicates primary key attributes

E-R Diagrams

1. Rectangles represent **entity sets**.
2. Diamonds represent **relationship sets**.
3. **Lines link** attributes to entity sets and entity sets to relationship sets.
4. Ellipses represent **attributes**
5. Double ellipses represent **multivalued attributes**.
6. Dashed ellipses denote **derived attributes**.
7. **Underline** indicates primary key attributes

Relationship Type & Relationship set

Relationship type:

- It represents the association between entity types.
- For example, 'Enrolled in' is a relationship type that exists between entity type Student and Course.
- In ER diagram, relationship type is represented by a **diamond** and connecting the entities with lines.



Keys

- Key is an attribute or collection of attributes that uniquely identifies an entity among entity set.
- There are different types of keys:
 1. Super key
 2. Candidate key
 3. Primary key
 4. Foreign key

Let's take an example to understand this:

Table: Employee

Emp_SSN	Emp_Number	Emp_Name
-----	-----	-----
123456789	226	Steve
999999321	227	Ajeet
888997212	228	Chaitanya
777778888	229	Robert

Keys..

Super key:

- Set of one or more attributes (columns), which can uniquely identify a row in a table.
- All the following sets of super key can uniquely identify a row of the employee table.
- {Emp_SSN}
- {Emp_Number}
- {Emp_SSN, Emp_Number}
- {Emp_SSN, Emp_Name}
- {Emp_SSN, Emp_Number, Emp_Name}
- {Emp_Number, Emp_Name}

Candidate key:

- It is a minimal super key with no redundant attributes.
- The following two set of super keys are chosen as candidate key.
- {Emp_SSN}
- {Emp_Number}

Primary key:

- It is selected from a set of candidate keys.
- It uniquely identifies the entity.
- This is done by database admin or database designer.
- We can say that either {Emp_SSN} or {Emp_Number} can be chosen as a primary key for the table Employee.

Foreign key:

- Foreign keys are the columns of a table that points to the candidate key of another table.

Example:

Stu_Id column in Course_enrollment table is a foreign key as it points to the primary key of the Student table.

Course_enrollment table:

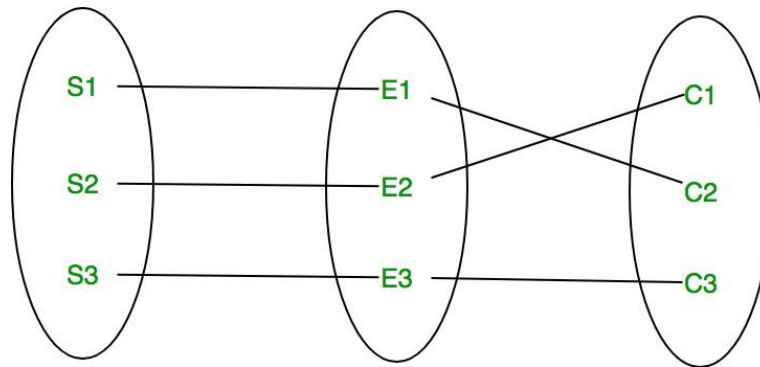
Course_Id	Stu_Id
C01	101
C02	102
C03	101
C05	102

Student table:

Stu_Id	Stu_Name	Stu_Age
101	Chaitanya	22
102	Arya	26
103	Bran	25
104	Jon	21

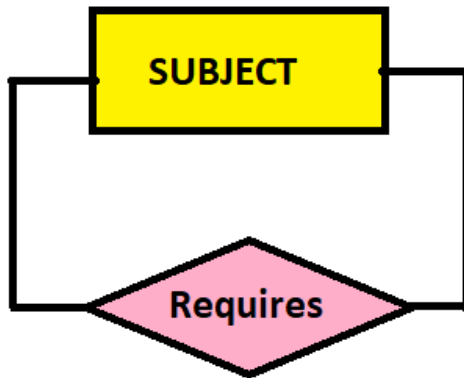
Relationship set:

A set of relationships of same type is known as relationship set. The following relationship set depicts S1 is enrolled in C2, S2 is enrolled in C1 and S3 is enrolled in C3.



Degree of relationship set

- The number of different entity sets participating in a relationship set is called as degree of a relationship set.



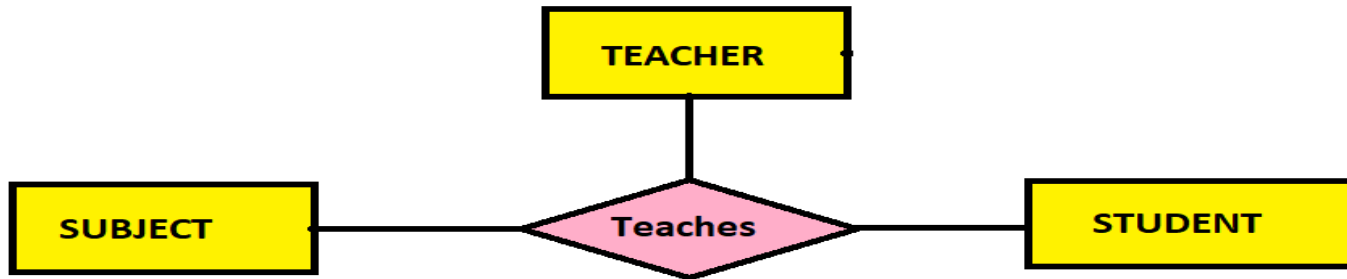
Unary relationship - Only ONE entity set participating in a relation



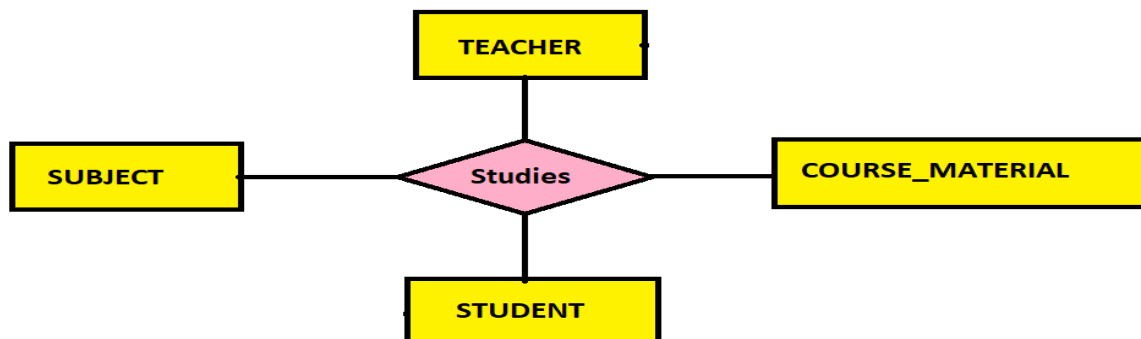
Binary relationship - TWO entities set participating in a relation

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- **Ternary:** exists when three entities are associated



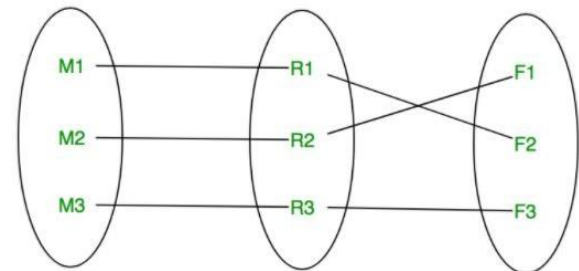
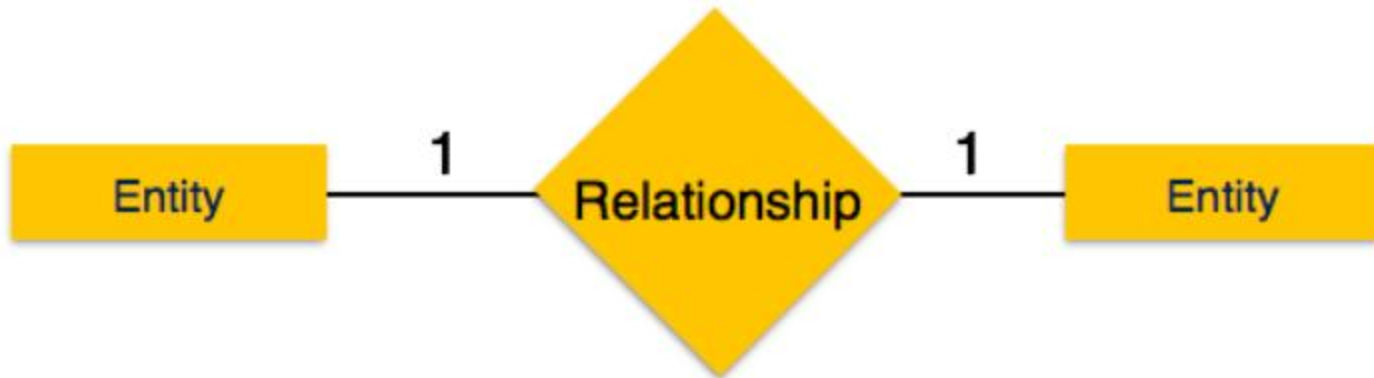
- **Quaternary :** four entities are associated



Cardinality

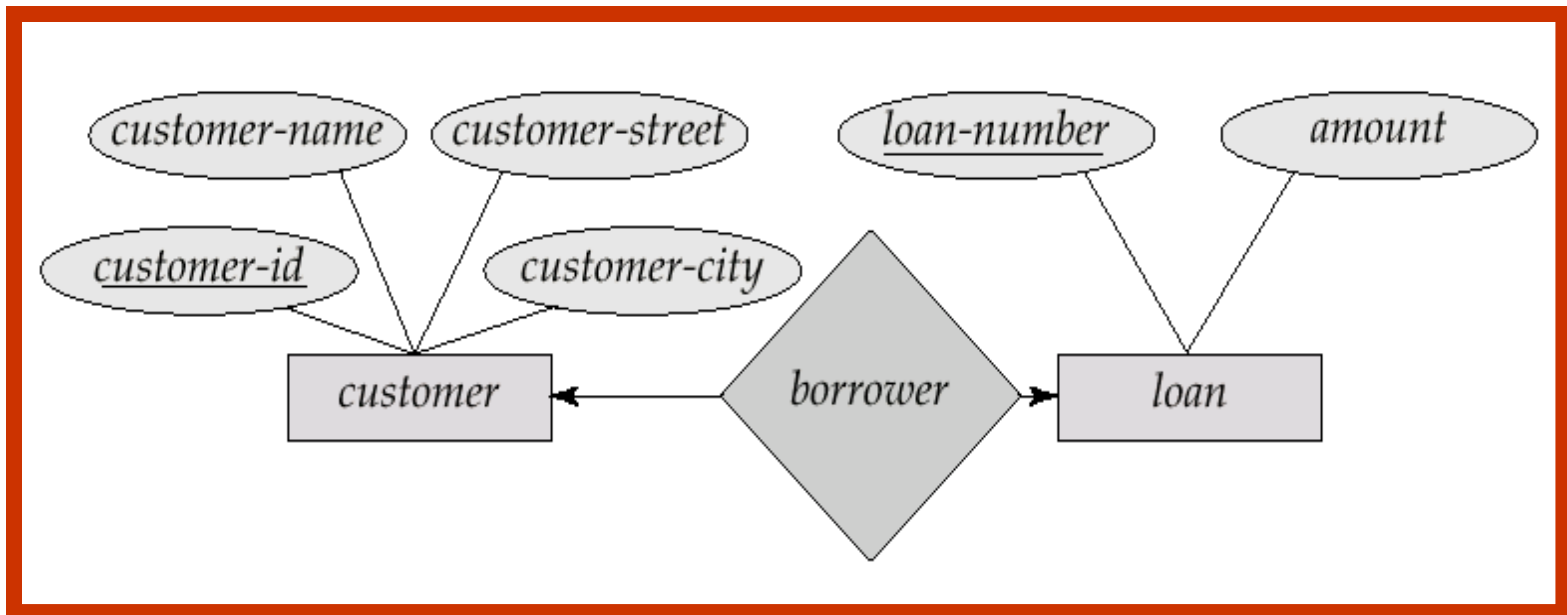
- The number of times an entity of an entity set participates in a relationship set is known as cardinality.
- Cardinality can be of different types:
 1. One to one
 2. Many to many
 3. One to many
 4. Many to one

- **One-to-one** – When only one instance of an entity is associated with the relationship, it is marked as '1:1'.



Cardinality Constraints

- We express cardinality constraints by drawing either a directed line ($\rightarrow$), signifying “one,” or an undirected line ($—$), signifying “many,” between the relationship set and the entity set.
- E.g.: One-to-one relationship
 - ☞ A customer is associated with at most one loan via the relationship *borrower*
 - ☞ A loan is associated with at most one customer via *borrower*



Participation Constraints

- **Total Participation** – Each entity is involved in the relationship. Total participation is represented by double lines.
- **Partial participation** – Not all entities are involved in the relationship. Partial participation is represented by single lines.



Partial participation



<u>EID</u>	Name	Qualification
1002	Ashwini Kunte	PhD
1003	Mita B	PhD
1007	Tanuja Sarode	PhD
1004	Madhuri Maam	PhD
1111	Ajay Vijay	ME

<u>EID</u>	<u>DID</u>
1002	40
1003	30
1007	10
1004	20

<u>DID</u>	Dname	Location
10	COMPS	NB1FLOOR
20	IT	NB8FLOOR
30	BIOMED	OB3FLOOR
40	EXTC	NB6FLOOR
50	CHEMICAL	OB5FLOOR

Total Participation

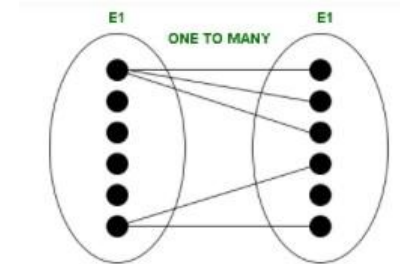
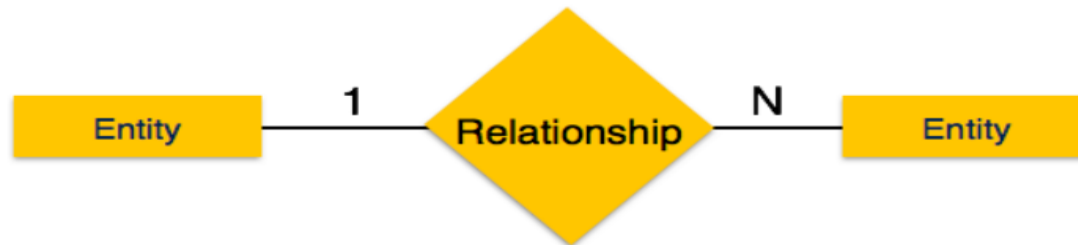


<u>EID</u>	Name	Qualification
1002	Ashwini Kunte	PhD
1003	Mita B	PhD
1007	Tanuja Sarode	PhD
1004	Madhuri Maam	PhD
1111	Ajay Vijay	ME

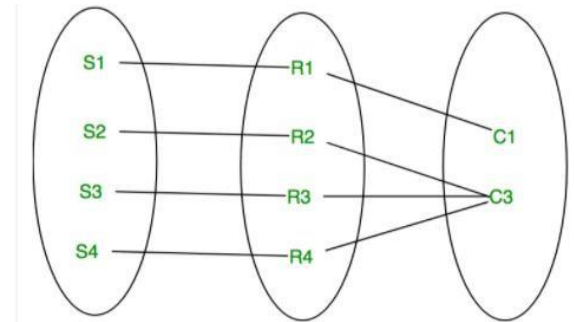
<u>EID</u>	<u>DID</u>
1002	40
1003	30
1007	10
1004	20
1111	50

<u>DID</u>	Dname	Location
10	COMPS	NB1FLOOR
20	IT	NB8FLOOR
30	BIOMED	OB3FLOOR
40	EXTC	NB6FLOOR
50	CHEMICAL	OB5FLOOR

- **One-to-many** – When more than one instance of an entity is associated with a relationship, it is marked as '1:N'. The following image reflects that only one instance of entity on the left and more than one instance of an entity on the right can be associated with the relationship.

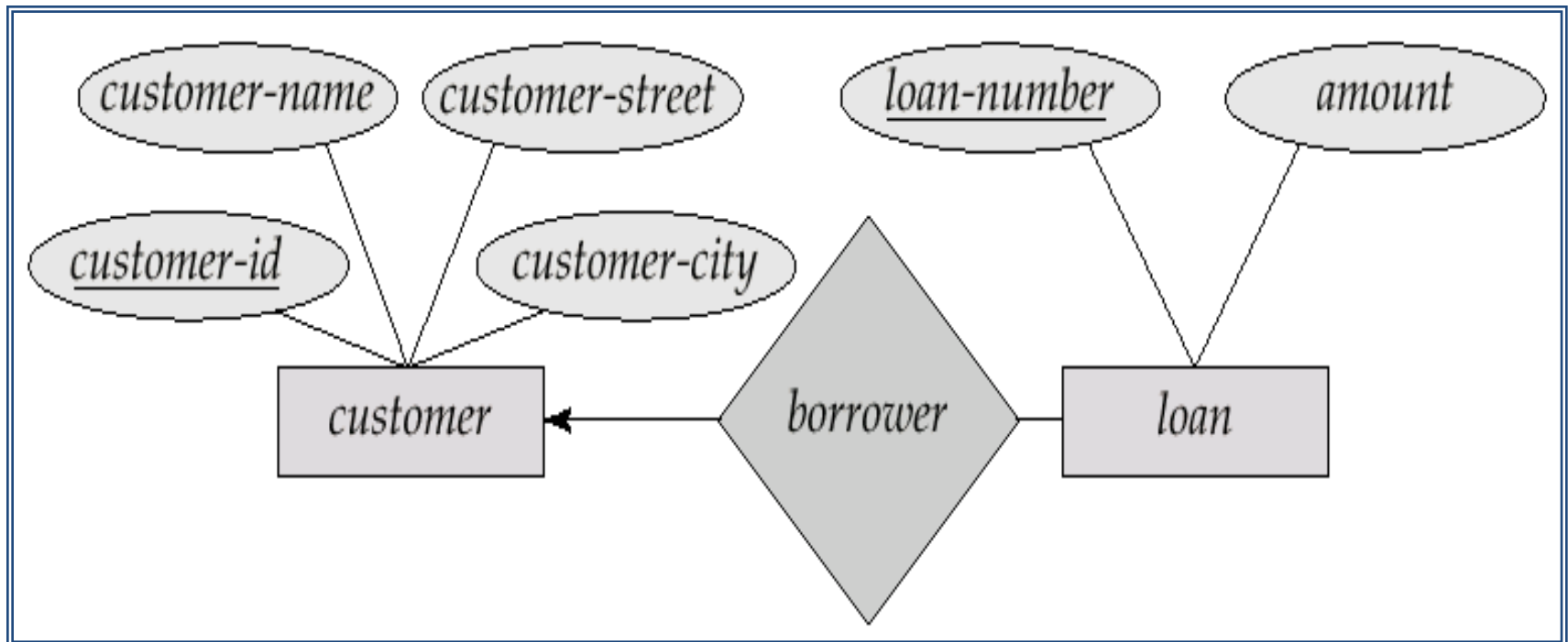


- **Many to one:**



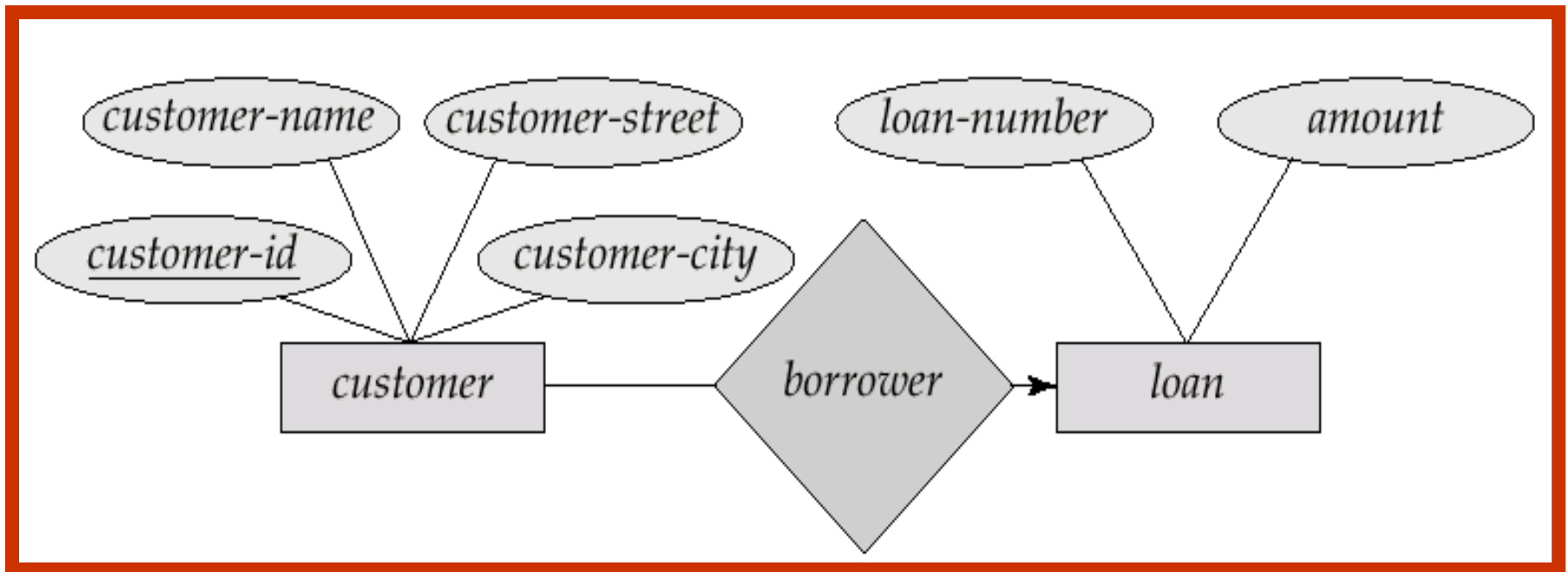
One-To-Many Relationship

- In the one-to-many relationship a loan is associated with at most one customer via *borrower*, a customer is associated with several (including 0) loans via *borrower*

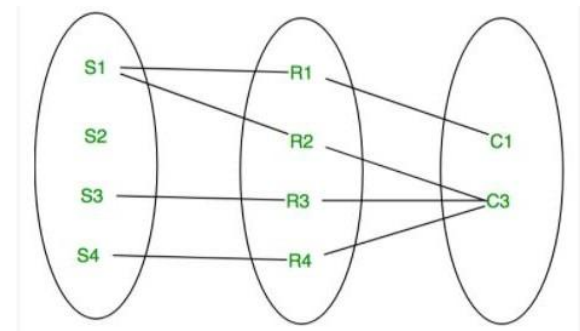


Many-To-One Relationships

- In a many-to-one relationship a loan is associated with several (including 0) customers via *borrower*, a customer is associated with at most one loan via *borrower*

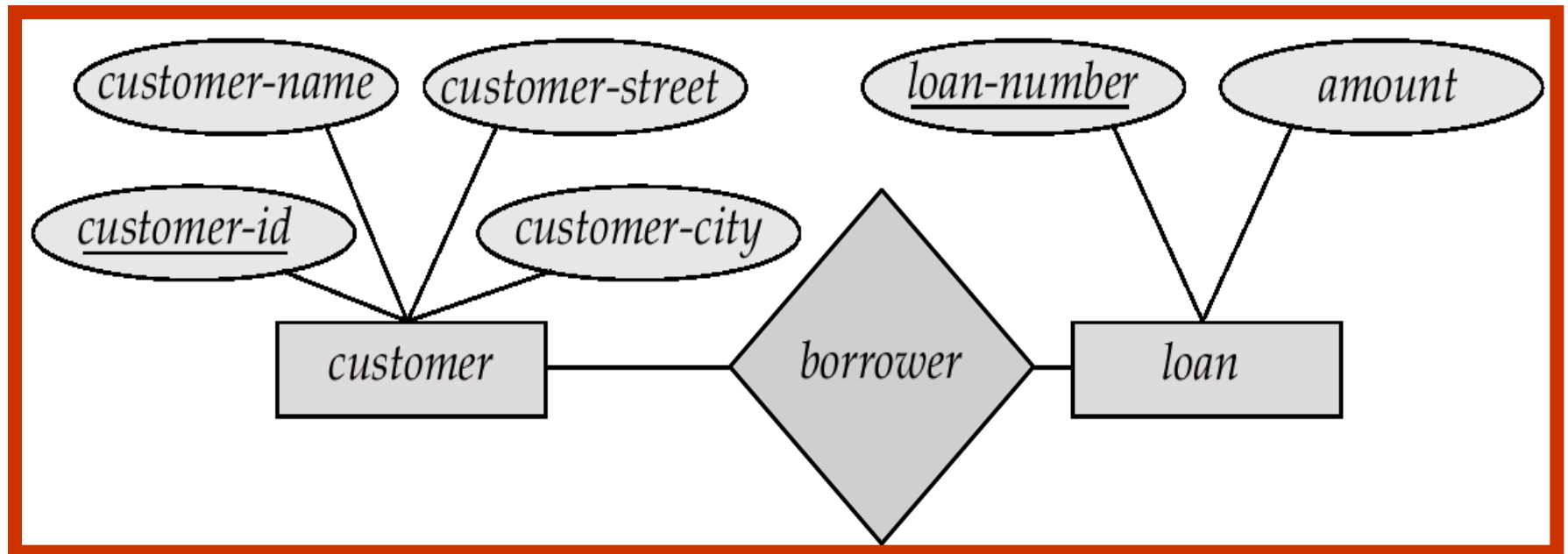


- **Many-to-many** – The following image reflects that more than one instance of an entity on the left and more than one instance of an entity on the right can be associated with the relationship. It depicts many-to-many relationship.



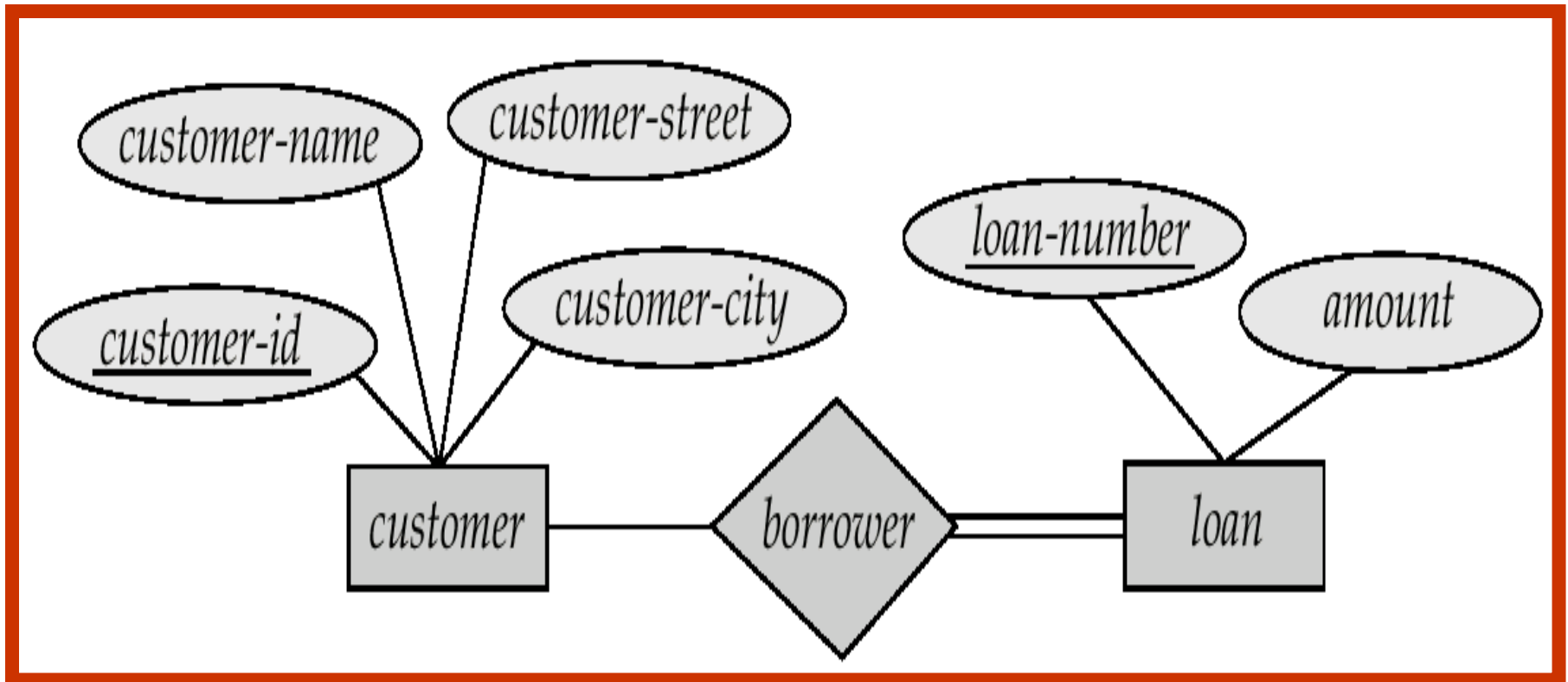
Many-To-Many Relationship

- A customer is associated with several (possibly 0) loans via borrower
- A loan is associated with several (possibly 0) customers via borrower



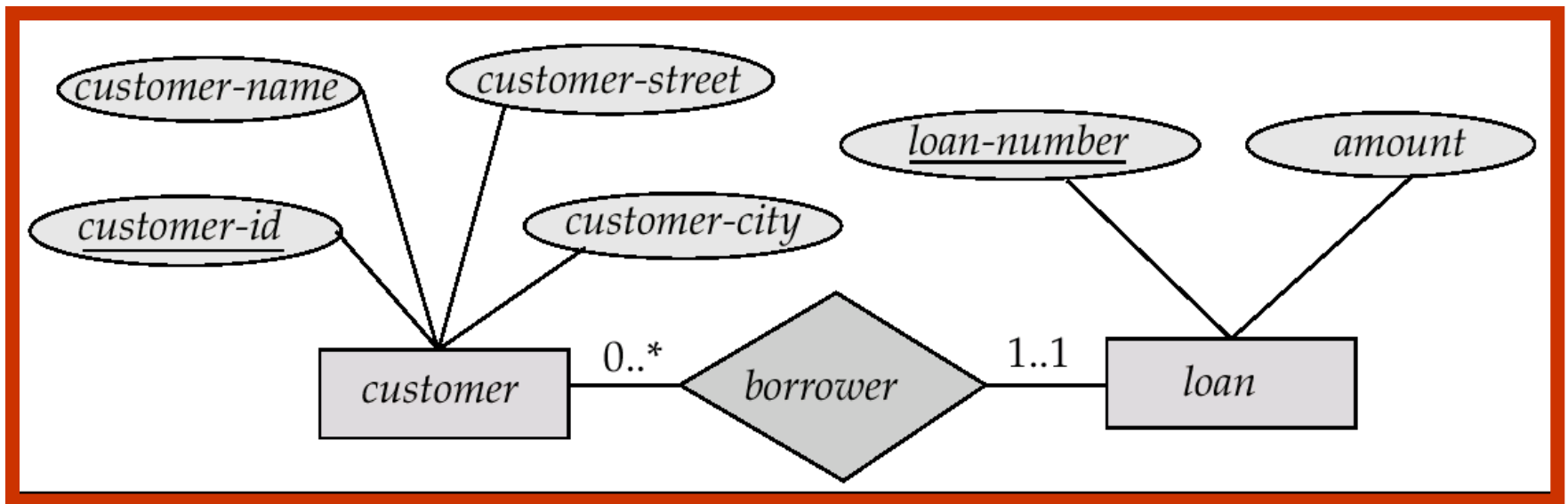
Participation of an Entity Set in a Relationship Set

- Total participation: indicated by double line
- Partial participation



Alternative Notation for Cardinality Limits

- Cardinality limits can also express participation constraints



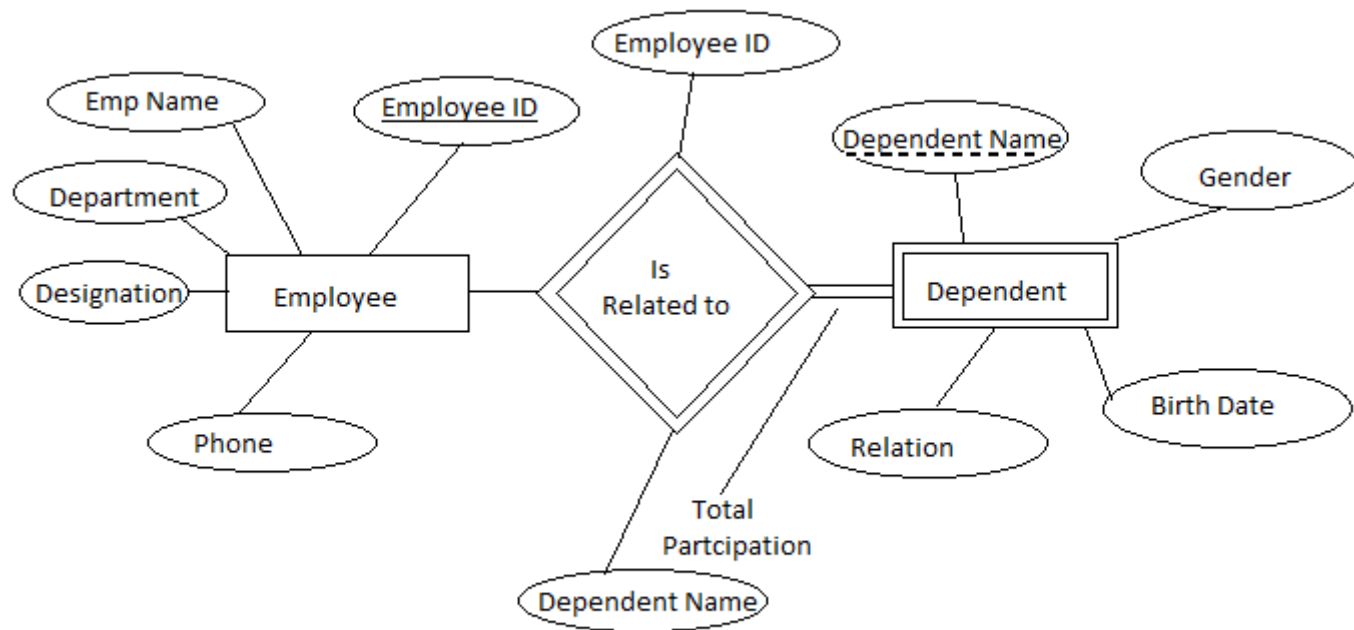
Entity set Types

- **Strong Entity** : Entities having its own attribute as primary keys are called **strong entity**.
E.g. : STUDENT has STUDENT_ID as primary key.
- **Weak Entity** : Entities which cannot form their own attribute as primary key are known **weak entities**.
- These entities will derive their primary keys from the combination of its attribute and primary key from its mapping entity.

Dependency

- Entities can be classified as
 - Strong / Parent /Owner/ Dominate
 - Weak entity type / existence- dependent/child/ Subordinate
- Existence- dependency means that if the existence of entity x depends on the existence of entity y, then x is said to be existence dependent on y
- A strong entity is not existence dependent whereas weak entities are

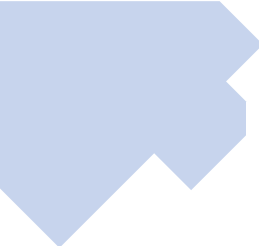
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Identifying Relationship set attributes are {Employee ID, Dependent Name}

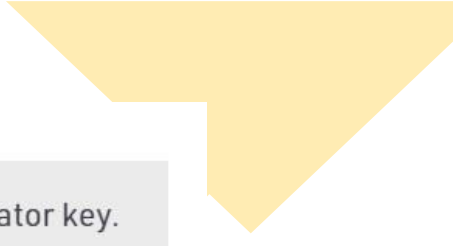
Comparison between Strong and Weak entity

Strong Entity Set	Weak Entity Set
it has its own primary key. It is represented by a rectangle. It contains a primary key represented by an underline. The member of strong entity set is called as dominant entity set. The Primary Key is one of its attributes which uniquely identifies its member. The relationship between two strong entity set is represent by a diamond symbol. The line connecting strong entity set with the relationship is single Total participation in the relationship may or may not exist.	It does not have sufficient attributes to form a primary Key on its own. It is represented by a double rectangle. It contains a Partial Key or discriminator represented by a dashed underline. The member of weak entity set is called as subordinate entity set. The Primary Key of weak entity set is a combination of partial Key and Primary Key of the strong entity set. The relationship between one strong and a weak entity set is represented by a double diamond sign. It is known as identifying relationship. The line connecting weak entity set with the identifying relationship is double. Total participation in the identifying relationship always exists.



S.NO Strong Entity

Weak Entity



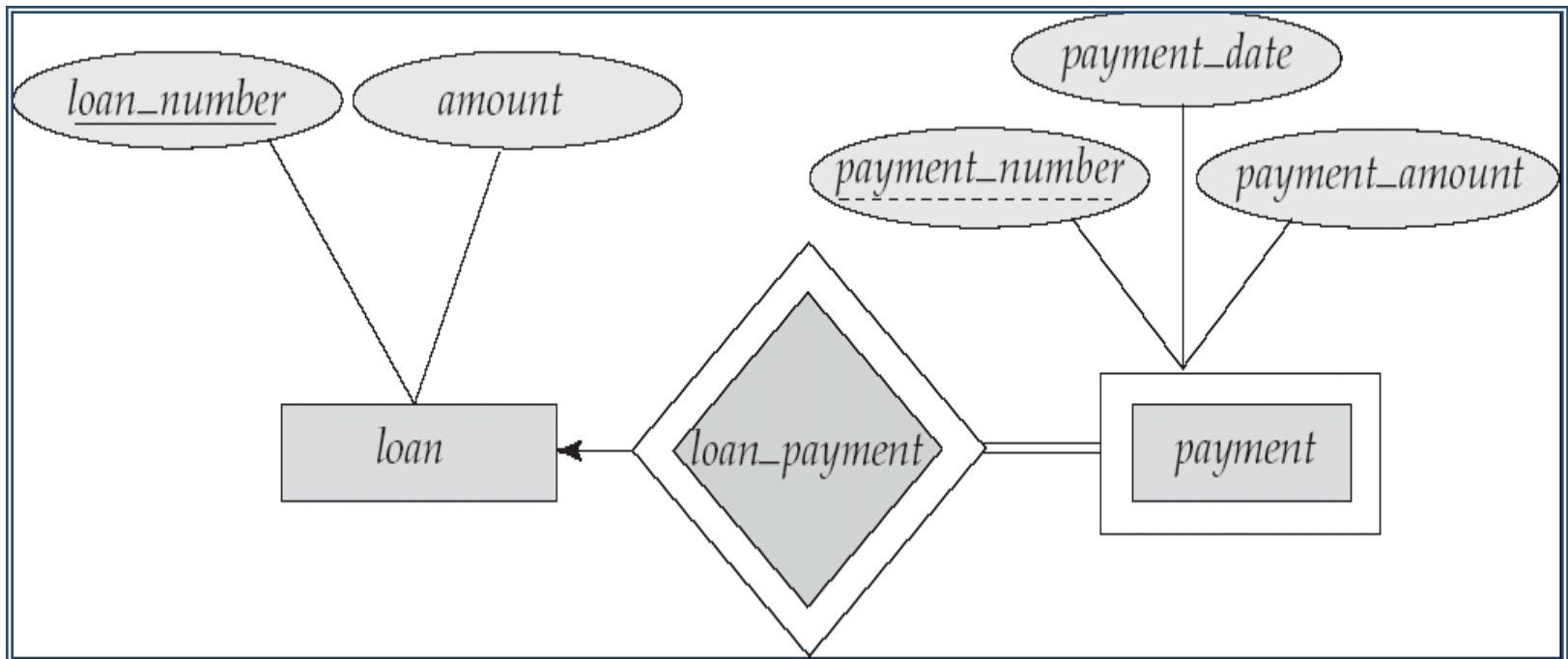
- | | |
|---|---|
| 1. Strong entity always has primary key. | While weak entity has partial discriminator key. |
| 2. Strong entity is not dependent of any other entity. | Weak entity is depend on strong entity. |
| 3. Strong entity is represented by single rectangle. | Weak entity is represented by double rectangle. |
| 4. Two strong entity's relationship is represented by single diamond. | While the relation between one strong and one weak entity is represented by double diamond. |
| 5. Strong entity have either total participation or not. | While weak entity always has total participation. |
- 

Weak Entity Sets

- The existence of a weak entity set depends on the existence of a **identifying entity set**
 - It must relate to the identifying entity set via a total, one-to-many relationship set from the identifying to the weak entity set
 - **Identifying relationship** depicted using a **double diamond**
- The **discriminator (or partial key)** of a weak entity set is the set of attributes that distinguishes among all the entities of a weak entity set.
- The primary key of a weak entity set is formed by the primary key of the strong entity set on which the weak entity set is existence dependent, plus the weak entity set's discriminator.

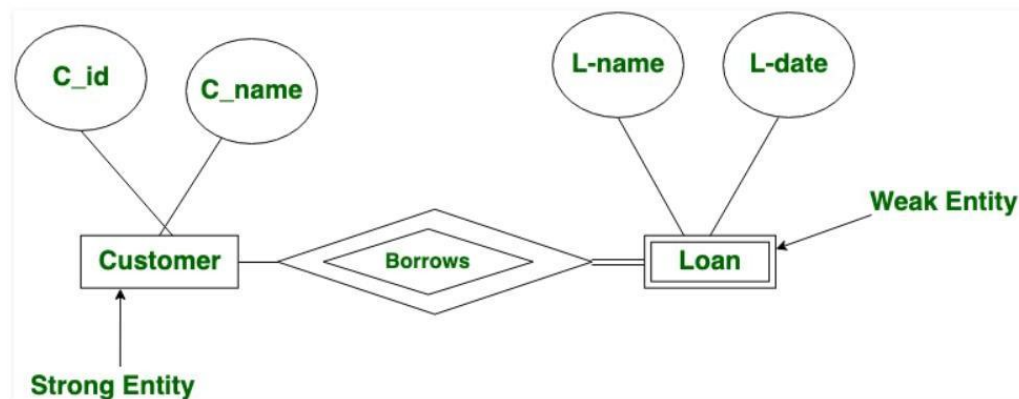
Weak Entity Sets (Cont.)

- We depict a weak entity set by double rectangles.
- We underline the discriminator of a weak entity set with a dashed line.
- `payment_number` – discriminator of the *payment* entity set
- Primary key for *payment* – (`loan_number`, `payment_number`)

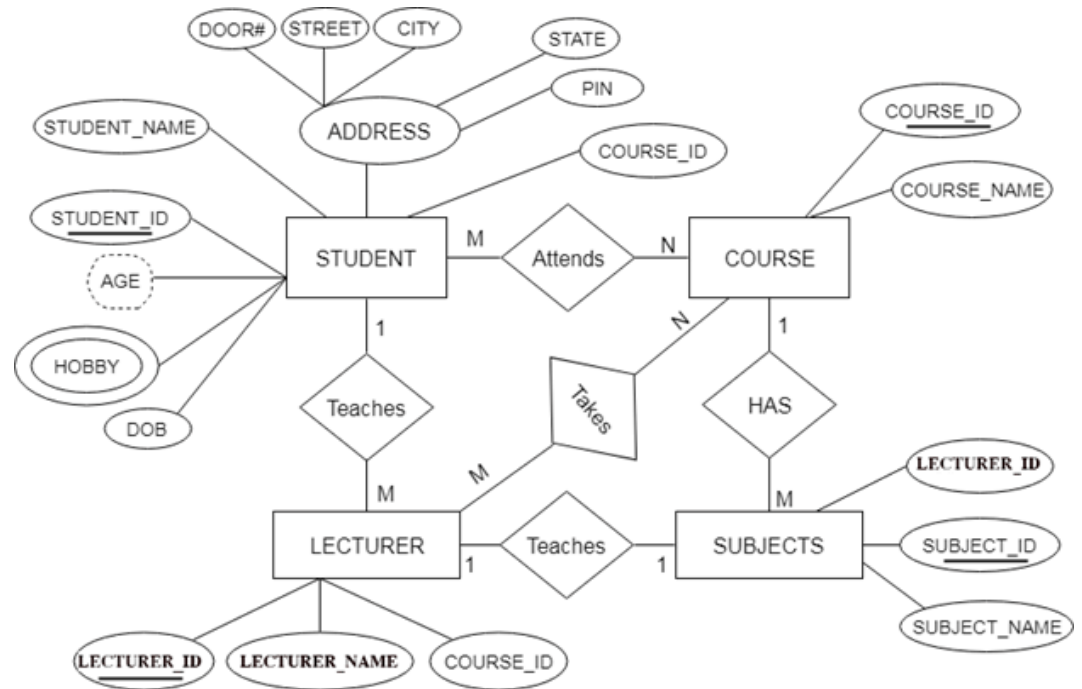


Weak entity:

- It is dependent on a strong entity to ensure its existence.
- Unlike a strong entity, a weak entity does not have any primary key.
- It instead has a partial discriminator key. A weak entity is represented by a **double rectangle**.
- The relation between one strong and one weak entity is represented by a **double diamond**.



College ER diagram

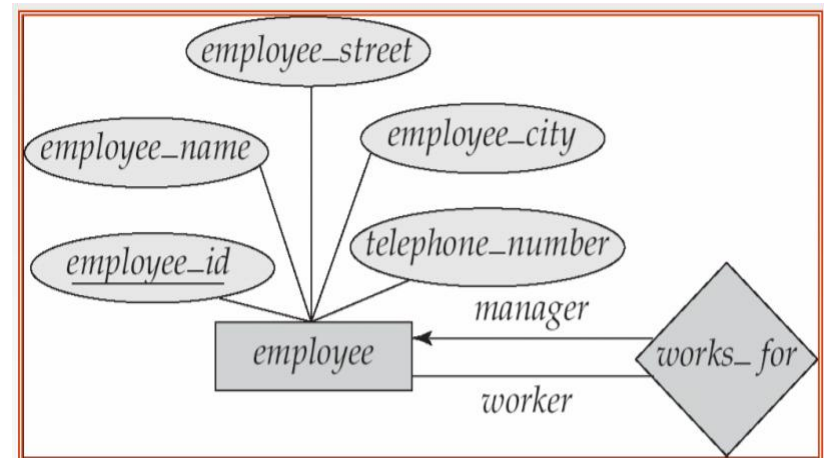


ER diagram with cardinality limits

- E-R diagrams also provide a way to indicate more complex constraints on the number of times each entity participates in relationships in a relationship set.
- An edge between an entity set and a binary relationship set can have an associated minimum and maximum cardinality, shown in the form **m..n**, where m is the minimum and n the maximum cardinality.

Roles

- Entity sets of a relationship need not be distinct
- The labels “manager” and “worker” are called **roles**; they specify how employee entities interact via the works_for relationship set.
- Roles are indicated in E-R diagrams by labeling the lines that connect diamonds to rectangles.
- Role labels are optional, and are used to clarify semantics of the relationship



Steps in ER Modelling

- Usually the following five steps are followed to generate ER models
 1. Identify the entity set.
 2. Identify the relevant attributes.
 3. Identify the prime attribute.
 4. Find relationships between entity set.
 5. Draw a complete ER model.

Example:

- Draw an ER Model for an University database application where
 - a) A University has many departments.
 - b) Each department has multiple instructors; one among them is the head of the department.
 - c) An instructor belongs to only one department.
 - d) Each department offers multiple courses, each of which is taught by a single instructor.
 - e) A student may enroll for many courses offered by different departments.

Example:

Step 1: Identify the entity set

- - From the given question, we can identify the following entity sets.
 1. DEPARTMENT
 2. COURSE
 3. INSTRUCTOR
 4. STUDENT
 - “Head of the department” is NOT an entity set; it is relationship between the INSTRUCTOR and DEPARTMENT entities.
different departments.

Step 2: Identify the relevant attributes

- For the **DEPARTMENT** entity set the relevant attributes are **Dept_No**, **Dept_Name** and **Location**.
- For the **COURSE** entity set the relevant attributes are **Course_No**, **Course_Name**, **Duration**, and **Pre-requisite**.
- For the **INSTRUCTOR** entity set the relevant attributes are **Inst_Id**, **Inst_Name**, **Room_No**, and **Telephone_No**.
- For the **STUDENT** entity set the relevant attributes are **Student_No**, **Student_Name**, and **Dob**.

Step 3: Identify the Prime (key) attribute

- **Dept_No** is the key attribute for **DEPARTMENT** entity set.
- **Course_No** is the key attribute for **COURSE** entity set.
- **Inst_Id** is the key attribute for **INSTRUCTOR** entity set.
- **Student_No** is the key attribute for **STUDENT** entity set.

Step 4: Identify the relationship between entity sets

1. Each department has multiple instructors and an instructor belongs to only one department.



2. Each department has multiple instructors; one among them is the head of the department.



3. Each department offers multiple courses.



4. Each department offers multiple courses, each of which is taught by a single instructor.



5. A student may enroll for many courses offered by different departments

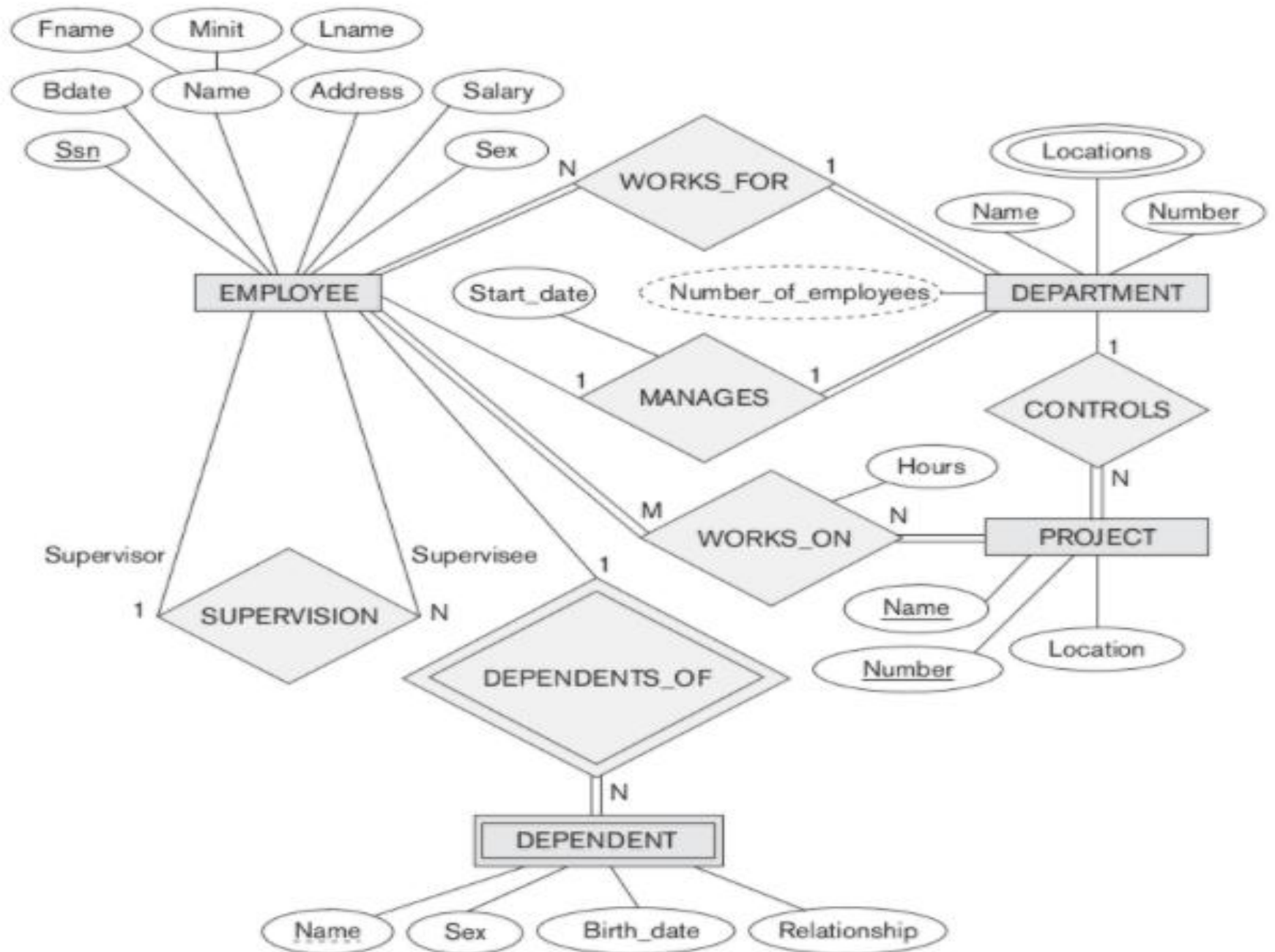


Step 5: Draw the complete E R Diagram



Sample Database Application

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's **name**, Social Security number,² address, salary, sex (gender), and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, sex, birth date, and relationship to the employee.



Library Management System

The Library Management System database keeps track of readers with the following considerations –

- The system keeps track of the staff with a single point authentication system comprising login Id and password.
- Staff maintains the book catalog with its ISBN, Book title, price(in INR), category(novel, general, story), edition, author Number and details.
- A publisher has publisher Id, Year when the book was published, and name of the book.
- Readers are registered with their user_id, email, name (first name, last name), Phone no (multiple entries allowed), communication address. The staff keeps track of readers.
- Readers can return/reserve books that stamps with issue date and return date. If not returned within the prescribed time period, it may have a due date too.
- Staff also generate reports that has readers id, registration no of report, book no and return/issue info.

